

**PUBLISHED**

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                          | Marks    |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(b)(ii) | <p><i>any <b>two</b> from:</i></p> <ol style="list-style-type: none"> <li>1 lactose is the <b>selection pressure</b> ;</li> <li>2 (<i>lac</i><sup>+</sup>), can metabolise / take up / AW, lactose (so survive)<br/><b>or</b><br/><i>lac</i><sup>-</sup>, cannot metabolise / take up / AW, lactose (so die) ;</li> <li>3 AVP ; e.g. <i>ref.</i> glucose has run out</li> </ol> | <b>2</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                      | Marks    |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a)     | <ol style="list-style-type: none"> <li>1 mutation ;</li> <li>2 horizontal transmission / conjugation / transformation / transduction ;</li> </ol>                                                                                                                                                                                           | <b>2</b> |
| 4(b)     | <p><i>idea of tracking, mutations / resistance genes / resistance alleles</i><br/><b>or</b><br/>identifies mechanisms of resistance<br/><b>or</b><br/>(informed) choice / production, of antibiotic ;</p>                                                                                                                                   | <b>1</b> |
| 4(c)     | <p><i>any <b>three</b> from:</i></p> <ol style="list-style-type: none"> <li>1 streptomycin is selection pressure ;</li> <li>2 bacteria with, mutation / resistance, survive / reproduce / have a selective advantage ; <b>ora</b></li> <li>3 allele / gene, for <b>resistance</b> passed on ;</li> <li>4 directional selection ;</li> </ol> | <b>3</b> |